Lumpy Skin Disease (LSD) in Cattle

Srinivas K, Santosh B Ghadage, Adhir Chaubal, Surya Prasad V, Swapna Reddy E and Madhanmohan M
Indian Immunologicals Limited, Hyderabad, Telangana and V U T D C, TANUVAS, Madurai, Tamil Nadu.

Introduction

Lumpy Skin Disease (LSD) caused by Lumpy Skin Disease Virus belongs to the *Pox viridae* family from the genus *Capripox virus* principally affecting the cattle *(Bos Taurus and Bos indicus)* and water buffaloes *(Bubalis bubalis)* (1). The first case of the disease was reported in 1929 from Zambia (2). The LSD virus shares a close antigenic resemblance with *Sheep Pox Virus* (SPV) and *Goat Pox Virus* (GPV) on the basis of morphology, serology, and the cytopathic effects in tissue cultures. Lumpy skin disease is an infectious, eruptive, occasionally fatal disease of cattle and water buffaloes characterized by nodules on the skin and other parts of the body. Secondary bacterial infections further complicate the condition. The incubation period is approximately 4 to 14 days and can be longer in natural infections. Other synonyms for the LSD are *Neethling Virus Disease, Exanthema Nodularis Bovis and Pseudo-Urticaria.* General provisions on the disease are defined in chapter 11.9 in OIE (3). There is only one serotype of LSD virus.

Transmission

Biting arthropod insect vectors are implicated in the transmission of virus. Grazing on shared pasture lands, close proximity to water sources, temperature, rainfall and wet humid summer months have been associated with outbreaks of LSD and are consistent with vector-borne transmission of virus. It is also possible to witness outbreaks of LSD throughout the year. Experimental studies found no evidence for direct transmission of the virus between animals, instead transmission is strongly believed to be via arthropod vectors (4). It is not clear whether vector borne transmission is biological or mechanical and movement of unvaccinated cattle from infected areas poses a major risk. LSD has been successfully transmitted experimentally via *Aedes aegypti* mosquitoes and common stable fly, *Stomoxys calcitrans.* The arthropod vectors, flies and ticks are thought to inoculate the virus into the skin of the cattle with virus circulating in the blood stream producing viremia and then further disseminating into distant cutaneous sites, GI tract, respiratory tract, reproductive tract and other internal organs. LSD is host specific and evidence of infection in species other than cattle and buffaloes is not reported. Skin lesions are the major source of infection: although the virus is released from body secretions and excretions including semen. Direct contact between animals does not play a major role in spread of the disease. Iatrogenic intra or inter herd transmission occur via contaminated needles during vaccination or if needles are not changed between animals. Eventually, affected animals clear the infection and there is no known carrier state for LSDV.

Disease distribution

In India, it was first reported from Odisha (Khairbani, Betnoti and Mayurbhanj) on 12th August 2019. Subsequently, the outbreaks were reported from other parts of Mayurbhanj, Bhadrak, Odisha. ICAR-National Institute of High Security Animal Diseases (ICAR-NIHSAD), Bhopal confirmed the virus identity by 16th November 2019. Epidemiological analysis and genetic characterization of Indian isolates revealed the closeness to South African isolates than to European isolates. During these outbreak episodes from Odisha, the researchers could detect the LSD virus genome by real time PCR from semen of naturally infected cattle (5). The disease is notifiable and classified under category A by OIE. The reasons for the sudden appearance of LSD in India could be due to climatic conditions conducive from globalization, importation of animals, disrupted livestock movement patterns associated with high insect activity and moment of people and goods.LSD cases are also reported from Chhattisgarh, Madhya Pradesh, Jharkhand, West Bengal, Assam, Andhra Pradesh, Telangana, Tamil Nadu, Maharashtra, Karnataka and Kerala. The disease is a classic example of a transboundary emerging disease and is not known to Europe till 2014 when it was first reported in Cyprus spreading further to different Balkan countries. Since 2019, several outbreaks of LSD have been reported from South Asia and East Asia. Outbreaks of LSD were reported in Africa till the year 2012 from where it began to spread to other parts of the world like Iran, Iraq, Jordan, Turkey, Russia, Balkan & Caucasus regions, Russia, Kazakhstan, Cyprus, Greece to France, middle east, west Asia, south and south east Asia (6). Recent outbreaks were reported from Bangladesh (July 2019) which was closely followed by Odisha outbreaks in the month of August 2019, China in August 2019 and Taipei in July 2020. The disease is endemic in most of Africa and few parts of Middle east and Turkey. The risk of transmission is very

high in hitherto unaffected countries. Cases of LSD are reported from Israel, Kuwait, Lebanon, Yemen, UAE, Bahrain, Oman, Saudi Arabia, Armenia, Azerbaijan, Bulgaria, Serbia, Kosovo, Macedonia and Montenegro (6, 7).

Clinical signs

Incubation period ranges from 2 to 5 weeks. The first sign of the disease is rise in temperature up to 41^{0}C followed by appearance of nodules over the skin in next 3 days. The disease is an acute infectious viral skin disease of cattle characterized by fever, various lumps or nodules on the skin covering all parts of the body, lymphadenitis and edema of the legs, udder, genitalia and brisket (8, 9). A subcutaneous injection of infected material produces a painful swelling, fever, lacrimation, nasal discharge and hyper salivation followed by the appearance of characteristic and distinctive circumscribed eruptions/nodules of 0.5 to 5 cms in diameter on the skin from a mild to severe form in susceptible cattle in 3 to 5 days. Their numbers may range from a few to several hundreds in a single affected animal.

The skin nodules contain a firm, creamy-gray or yellow mass of tissue. The nodules or lumps are painful, involve the epidermis, dermis, and subcutaneous tissue. They are firm in nature, raised, round or circumscribed, may disappear and start again in other parts of the body or persist, the overlying skin sometimes necrosed, and sloughs to leave large open sores. The virus is present in blood, nasal, lachrymal secretions, saliva and semen. The skin lesions are often accompanied by oral, nasal and ocular discharge, lethargy, anorexia, a rapid drop in milk production. After 1–2 weeks the skin nodules become necrotic and slough off leaving ulcers and scars over the skin. The course of the disease lasts from 5 to 8 weeks.

In rare cases like in Foot and Mouth Disease (FMD) affected cattle, nodules may be found on muzzle, nasal and oral mucus membranes. Though infection rates are higher in affected cattle, the mortality is less. There is no evidence of human infection. Experimentally, buffalo, sheep, goats, and giraffes can be infected with mild clinical signs and are believed to be the reservoirs. The disease is more sever in young cattle though it is known to occur in all age groups of cattle. Pneumonia caused by the virus itself or secondary bacterial infections, and mastitis are common complications. Skin lesions in the legs and on top of the joints lead to deep subcutaneous infections complicated by lameness. When an animal with multiple skin lesions is sent to a slaughterhouse, subcutaneous lesions are clearly visible after the animal is skinned. In a postmortem examination, pox lesions can be found in digestive and respiratory tracts and on the surfaces of internal organs. A few pictures of animals affected with LSD are shown in **figures 1 to 9.**

Economic losses

The disease inflicts significant economic losses due to reduction in body condition, milk loss, poor quality of hides, infertility problems in both males and females, abortions and serious trade restrictions and controls. Course of the disease is usually between 5 to 8 weeks. Full recovery of affected cattle takes several months and may prolong where secondary bacterial infection occurs and obstruction of respiratory passages by the lesions may lead to deaths in some cases especially in calves.

Morbidity and mortality rates

Morbidity and mortality in LSD outbreaks are lower than sheep pox and goat pox. Morbidity varies between 2 to 45% while the mortality is usually less than 10% (8). The resistance to the disease is influenced by immune status, age and breed rather than the virulence of the virus. Reports from LSD outbreaks in Middle East and Europe reveal a morbidity of 9%–26%, and mortality from 0.5% to 2%. Fifteen years' data from Uganda shows a morbidity of 4.77% and a mortality of 0.03 % indicating a lower prevalence. Pure bred and cross bred animals are more prone to the infection and younger ones suffer more severe form of the disease (9, 10).

Diagnosis

The disease can be diagnosed by the appearance of distinct cutaneous lesions which can also be seen in bovine herpes virus known as pseudo lumpy skin disease. In bovines, skin nodules are also caused by Dermatophilus congolensis. LSD virus shares close resemblance with SPV and GPV and grows on tissue cultures of ovine, bovine, monkey and rabbit origin producing cytopathic changes and intracytoplasmic inclusion bodies. The virus can also be propagated in chick embryos and eggs. Immunofluorescent or immunoperoxidase staining are largely helpful in specific identification of the agent. Virus isolation can also be attempted in skin biopsy samples. Virus Neutralization Test, ELISA, PCR and real time PCR techniques are helpful in detection of the antigens. International reference laboratories for LSD are located in Onderstepoort, South Africa and The Pirbright Institute, U. K (6). In India, suspected clinical samples are advised to be forwarded to ICAR- NIHSAD, Bhopal for further diagnosis.

Treatment

Antibiotics are advocated for the control of secondary bacterial infections and symptomatic

treatment with help from Veterinarian is useful in reducing the clinical signs. Administration of anti-pyretics, anti-inflammatory and anti-histamines to reduce the suffering coupled with administration of multi vitamins, antiseptics and fly repellants. Ethnoveterinary medicine is also useful to treat the infected animals. Feeding soft gruel, feed and fodder is advised in affected animals. LSDV is very resistant to physical and chemical agents, can persist in necrotic skin lesions for more than 30 days. LSDV is killed by 2% sodium hydroxide, 4% sodium carbonate, 2% chloroform and formalin, 2% phenol, iodine and quaternary ammonium compounds at 1: 33 dilutions and 0.5% respectively. Isolation and quarantine measures are helpful but to a lesser extent. The virus is susceptible to higher temperatures and gets killed at 55^0C for 2 hours and at65^0C in 30 minutes.

Control by Vaccination

LSD is a vector borne disease which makes it difficult to predict and manage the spread and the best solution for the protection of animals is vaccination. Prophylactic vaccination using a live attenuated homologous Neethling strain vaccine administered subcutaneously has been in use in many countries and provides a robust immunity. The minimum recommended field dose (11) of the South African Neethling strain vaccines is $10^{3.5}$ $TCID^{50}$ although the minimum protective dose is $10^{2.0}$ $TCID^{50.}$

Heterologous protection by Sheep pox vaccine @ 0.5 ml by intradermal route is used in Egypt with a Romanian strain. A vaccine strain was developed in Kenya using a strain of sheep pox virus which had been passaged 16 times in lamb testis cells. A complete cross immunity had been demonstrated to exist between strains of sheep pox and LSD. This strain was used for many years at an immunizing dose of $10^{3.5}$ $TCID^{50}$. Capripox viruses are cross-reactive within the genus. Hence, it is possible to protect cattle against LSD using strains of Capripox virus derived from sheep or goats (12). However, it is recommended to carry out clinical trials and suitability studies and challenge studies using the most susceptible breeds, prior to introducing vaccine strains not usually used in cattle. Vaccination of cattle over 6 months of age offer protection. Goat pox vaccine is also used in a few countries as some cross protection is observed. Large scale vaccination combined with other appropriate control measures are the most effective way of limiting the spread and economic impact due to LSD. We as a country are fully braced up to face such emerging challenges as transboundary animal diseases are likely to hit in future too as recently we witnessed emergence of African Swine Fever (ASF) outbreaks in Assam, Arunachal Pradesh and Meghalaya in 2020. The ASF disease was also confirmed by ICAR-NIHSAD, Bhopal in May, 2020.

References

1. Tuppurainen, E.S.M. and Oura, C.A.L. (2012). Review: Lumpy Skin Disease: An emerging threat to Europe, the Middle East and Asia. Transbound. Emerg. Dis. 59: 40.
2. Haig, D. A. (1957). Lumpy skin disease. Bull. Epizoot. Dis. Afr.5: 421.
3. OIE, Terrestrial Animal Health Code. (2018): Chapter 11.9, Infection with Lumpy Skin Disease: General Provisions.
4. Tupparainen, E.S.M., Venter, E.H., Coetzer, J.A. and Bell-Sakyl, L. (2015): Lumpy skin disease: attempted propagation in tick cell lines and presence of viral DNA in field ticks collected from naturally-infected cattle. Ticks Tick Borne Dis. 6: 134.
5. Shashi B. Sudhakar , Niranjan Mishra , Semmannan Kalaiyarasu , Sandeep K. Jhade , Divakar Hemadri , Richa Sood , Gopal C. Bal , Mihir K. Nayak , Sunil K. Pradhan and Vijendra P. Singh. (2019): Lumpy skin disease (LSD) outbreaks in cattle in Odisha state, India in August 2019: Epidemiological features and molecular studies: Transboundary and emerging diseases.
6. OIE (World Organization for Animal Health). (2017). Lumpy Skin Disease. World Animal Health Information Database: http://www.oie.int/wahis_2/public/wahid.php/Wahidhome/Home.
7. FAO / EMPRESS, Publication on Lumpy Skin Disease. (2017) – Special issue No 47.
8. Coetzer, J.A.W. (2004). Lumpy skin disease. In J.A.W. Coetzer & R.C. Tustin (eds.) Infectious Diseases ofLivestock, 2nd ed. pp. 1268. Cape Town, OUP Southern Africa.
9. Hawsar Yasin Abdulqa, Heshu Sulaiman Rahman, Hiewa Othman Dyary and Hemn Hasann Othman. (2016): Lumpy Skin Disease: a mini review. Reprod. Immunology Open access. Vol 1. (4): 25.
10. OIE. (2010). Terrestrial Manual of Lumpy Skin Disease, Chapter 2.4.14. (Version adopted by the World Assembly of Delegates of the OIE in May 2010, OIE, Paris.).
11. Mathis, E., Vandenbussche, F., Haegeman, A., King, A., Nthangeni, B., Potgieter, C., Maartens, L., Van Borm, S., and De Clercq, K. (2016): Complete Genome Sequences of the Neethling-Like Lumpy Skin Disease Virus

Strains Obtained Directly from Three Commercial Live Attenuated Vaccines. Genome Announc. 4: (6).

12. Capstick, P.B. and Coakley, W. (1961). Protection of cattle against lumpy skin disease. Trials with a vaccine against Neethling type infection. Res. Vet. Sci. 2: 362.



Fig.1 : Exposed skin lesion in cattle



Fig. 2 : Nodules on the head of a calf.



Fig. 3 : Numerous nodules over skin of a cattle



Fig. 4 : Nodules all over the body of a cattle.



Fig.5 : Cutaneous nodules on the muzzle of a cattle.



Fig. 6 : Nodules on the dewlap region with swellings.



Fig.7 : Extensive nodules all over the body in a calf.



Fig.8 : Skin nodules in a cow



Fig.9 : Skin nodules in a bullock